

Chapter 4

The Quantum Mechanics of Light

Learning goals:

- (1) Describe what the quantum mystery is and show that it cannot be explained with classical physics alone.
- (2) Use Dirac notation to be able to summarize experiments in terms of state preparation and measurement.
- (3) Analyze delayed-choice experiments and verify that the results always come from what the state is at the time of measurement.
- (4) Discuss the difference between entangled and product states and why one shows interference effects more easily than the other.

Material to review:

- (a) Dirac notation.
- (b) Complex arithmetic.
- (c) Frequencies of visible light.
- (d) Single-photon sources and detectors.

In this chapter, we discuss the quantum mechanics of light. The approach is based on a methodology developed by Richard Feynman when he visited New Zealand and gave a series of four lectures on quantum mechanics in 1979. The original videos are available at <http://www.vega.org.uk/video/subseries/8> and the transcripts have been summarized into a popular book, *QED: The Strange Theory of Light and Matter* [1]. You can see an image of Richard Feynman in Fig. 4.1.

4.1 Feynman's Model for Light

Richard Feynman is one of the most well-known physicists in America. He created a novel way of approaching quantum mechanics by using what is called a path integral.



Fig. 4.1: Richard Feynman, a famous American physicist who loved to popularize science from a photo taken in 1958. He developed the path-integral approach to quantum mechanics and promoted the use of Feynman diagrams to describe perturbation theory in particle physics. [Image credit: AIP Emilio Segrè Visual Archives, Segrè Collection. Image converted to look like a stencil by the authors.]

This is an approach that assigns a complex number, determined by the action, for any path corresponding to an event that one wants to describe. By summing over all possible paths, one constructs the quantum prediction or probability amplitude for the event to occur.

In general, Feynman's approach is much more difficult to use for conventional quantum-mechanical problems than virtually any other method. It shines in particle physics, but requires significant mathematical tools to be able to use it. In the late 1970's, Feynman discovered a way to simplify the approach, but maintain its accuracy, for a class of problems in quantum mechanics. This allows these problems to be solved both accurately and efficiently and provides a strong physical intuition for how quantum phenomena occur. It is this version of his approach that we will discuss here.

We begin by recalling what a photon is. We learned in Chap. 1 that single photon experiments can be performed if one uses single-photon sources, and that these photons can be measured once and only once for any experimental run. Feynman's approach models these experiments in the following way. First, we identify the different events that occur that we are trying to describe. All of our events start with a photon leaving a source and they end with a photon being detected in a detector. If we have multiple detectors, we can have multiple events. Then, for each event, we identify the alternative ways the event can occur. These are the paths we will use for the approximation to the path integral. Each alternative way (path) is assigned a complex number (also referred to as an arrow). We add all the complex numbers

for each alternative way to find a final probability amplitude. This amplitude is then squared to determine the probability for the event to occur. If we have multiple events, we do the same analysis for each event.

We still need a set of rules to determine the complex number for each alternative way. The photon starts from the source with a unit amplitude and zero phase. As it moves from the source to the detector, the phase evolves via $e^{-i\omega t}$, where ω is the angular frequency of the photon (determined by its color). Note that the photon is not precisely monochromatic, as it involves a superposition of nearby frequencies to have a reasonably well-defined location, but it is nearly monochromatic, so we use its average frequency to describe the motion. We will discuss the other rules as we need them.

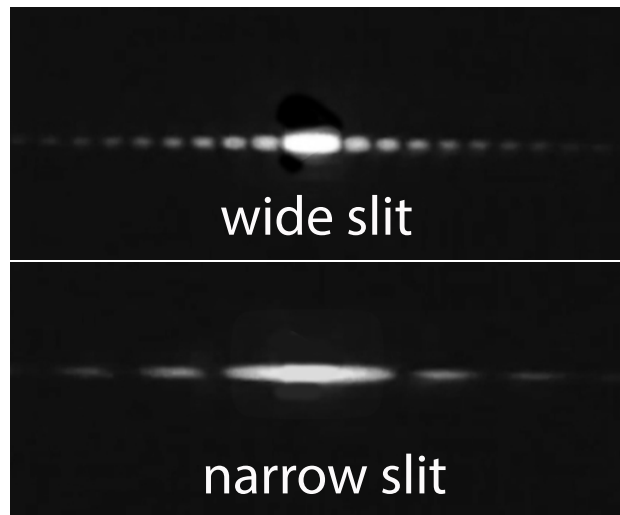


Fig. 4.2: Experimental images of single slit diffraction from a wide slit (top) and a narrow slit (bottom). Notice how the central region is quite bright and the sidebands get progressively dimmer as one moves away from the center. These images were taken from a video on single slit diffraction made by the technical services group of the Massachusetts Institute of Technology [used with permission © Massachusetts Institute of Technology].

The first problem we want to investigate is what is called single-slit diffraction. This is the pattern that appears on a far-away screen when light is shone through a screen that has a thin slit in it. In Fig. 4.2, we show the results for a single-slit diffraction pattern with a wide and narrow slit. We first describe how to explain the behavior classically and then show the similarities with the quantum treatment. We also point out the differences.

Consider a slit of width d that light shines through. We separate the slit into 12 sections and follow the light for each path, as shown in Fig. 4.3. For the light that

goes straight forward to the far away screen, we can see, everything is in sync. As the light wave oscillates up and down it does so together for all twelve pieces. This means at its peak, each section will have the same amplitude of A_0 . Each section occupies a smaller width of $\Delta d = \frac{d}{12}$, so the total amplitude will be $12 \times A_0 \times \Delta d = A_0 d$. This is the same as integrating the amplitude from 0 to d .

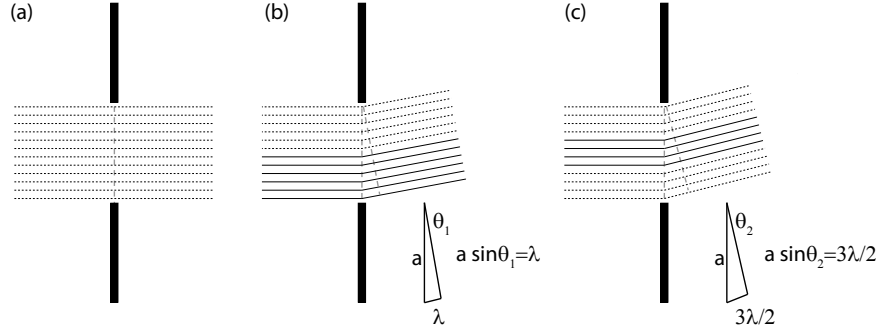


Fig. 4.3: Geometry for the classical calculation of the diffraction pattern on a screen. (a) In the pattern that goes through at 0° , all paths (dotted lines) add with exactly the same phase, and we get a large peak. (b) At the angle θ_1 , where the phase of the last path corresponds to a distance λ longer than the first path, the dashed lines add positively while the solids negatively, and each dashed has a solid partner that cancels, so this angle leads to a node in the pattern. (c) At the angle θ_2 , we have a difference of the distance of $\frac{3}{2}\lambda$ from the first to the last. Here the first set of dashed and solid lines cancel leaving just the second dashed set to give us a small peak.

Now consider an angle where the light does not go straight through. We need to focus on the light beam profile, which lies along the line perpendicular to the direction it is moving towards the screen (remember, light is a transverse wave). Now, we can see as we go from top to bottom, the light travels ever increasing paths. If the extra distance traveled by the bottom path versus the top path is one wavelength of the light, then the the profile along the wavefront looks like a sine curve, and we can see every light path with a given amplitude has a partner with the exact opposite amplitude, so they cancel. The net result is the light cancels itself out, and no light reaches the screen in this direction.

Now, increasing the angle further, we reach the point where we have one and a half wavelengths along the wavefront. As occurred for a full wavelength, we always find canceling partners for the first wavelength, so that part contributes nothing, but for the final half wavelength, the results do add up. To estimate how much smaller the amplitude is here, we compute the ratio

$$\text{Relative amplitude} = \frac{\int_0^{3\pi} dx \sin x}{\int_0^{3\pi} dx 1} = \frac{2}{3\pi}. \quad (4.1)$$

This estimate is simply comparing the net amplitude for a system with the partial cancellation ($\sin x$), to what it would be if everything added in phase (1). Since the intensity is proportional to the square of the net amplitude of the light, we find the maximum of the first side band should be $\frac{4}{9\pi^2} \approx \frac{1}{22}$. We expect the first sideband peak to be about 20 times smaller in height than the main peak.

Proceeding further, we obtain a zero, when the difference is two wavelengths. The next sideband peak will have an amplitude of $\frac{2}{5\pi}$ and so will be 62 times smaller than the main peak, but only about three times smaller than the first side band. This is precisely what the intensity profile looks like for the single-slit diffraction.

So, a single-slit diffraction has a very bright middle peak, and dim sidebands that fade off as one goes further away from the center. If one does not have enough contrast to see them, the sidebands can be hard to see, even the first one.

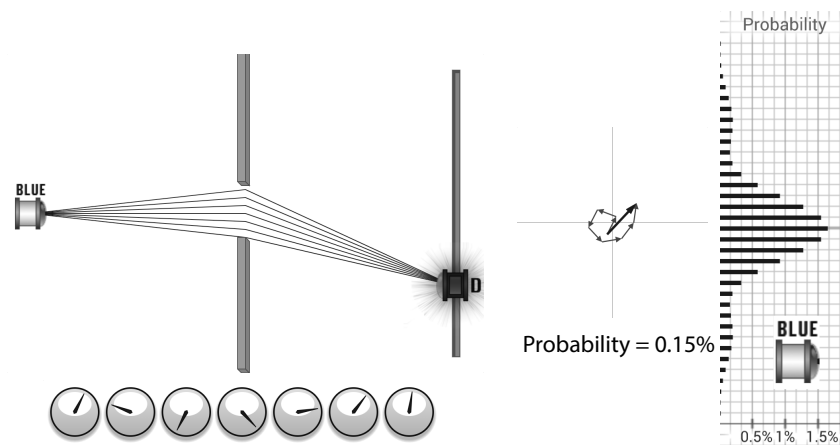


Fig. 4.4: Methodology for the quantum approach, here shown with seven paths instead of the twelve discussed in the text.

We just finished the classical analysis. Now, how does the quantum one work? Well, we first determine the event, which is a photon leaves the source, goes through the slit, and is detected on the far away screen at a particular location. A schematic for how this will work is given in Fig. 4.4 where we show a simpler example with seven paths. The clocks below show the directions of the arrows for each alternative way. The plot to the right shows what the pattern is after looking at all locations on the screen.

For the case of light going straight through, the time for each of the twelve paths we will use is identical, so they all line up in exactly the same direction, and when we add them together, we will find we obtain a length of 12 times the length of a single path. This will be our reference, and we will consider other cases relative to this.

For the case where the total path difference is one wavelength, we will find the arrows are equally distributed around the unit circle. If we hook them together head to tail, we see they exactly cancel when added together. This is consistent with seeing a dark spot at that angle.

If we have one and a half wavelengths, the first two thirds of the arrows are uniformly distributed around the unit circle, while the last third are uniformly distributed about the half circle. The first two thirds cancel when added together, and the second two thirds do add, but the final result is less than four times the length of a single arrow. The net, will be a result similar to what we calculated classically. One can continue this analysis for the other cases, and it should be clear that if we use enough paths, we will find the same predictions as the classical case.

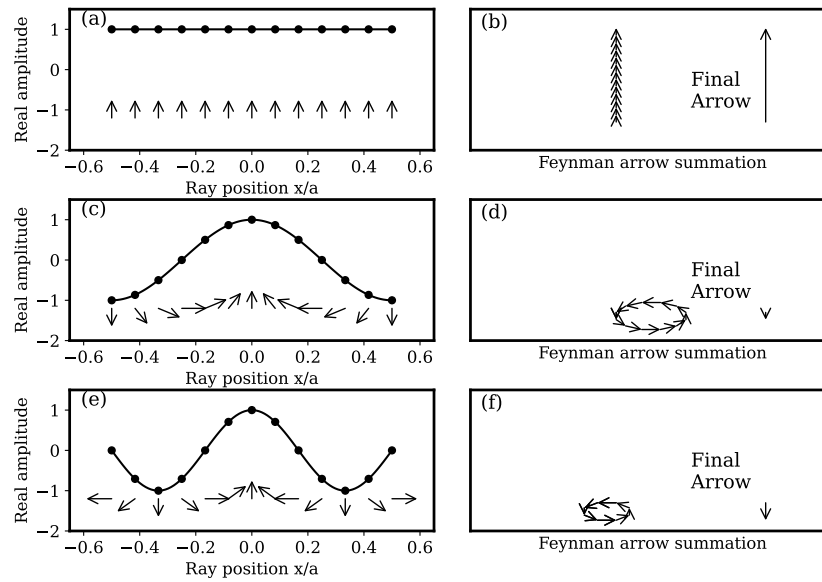


Fig. 4.5: Comparison of the classical and quantum way to calculate the diffraction patterns. In the left panels [(a), (c), (e)], we show the imaginary part of the wavefront at an instant of time, with the Feynman arrows below for each of the twelve rays. On the right [(b), (d), (f)], we show how the arrows line head to tail and what the sum looks like. For the straight paths $\theta = 0$, they arrows all line up. When we are at the angle θ_1 , the arrows all cancel. When we are at θ_2 , we wind around the ellipse one and a half times, ending with a small arrow.

We illustrate these two ways of calculating the diffraction pattern in Fig. 4.5. On the left, we show the classical wavefront at an instant in time and the Feynman arrows underneath for the twelve paths that we took. On the right, we show the Feynman summation of the arrows. One can see that both approaches will give the same results. Note that we have simplified the analysis for the classical case by plotting

the imaginary part of the wavefront rather than the full complex number (which is shown above with the Feynman arrows.

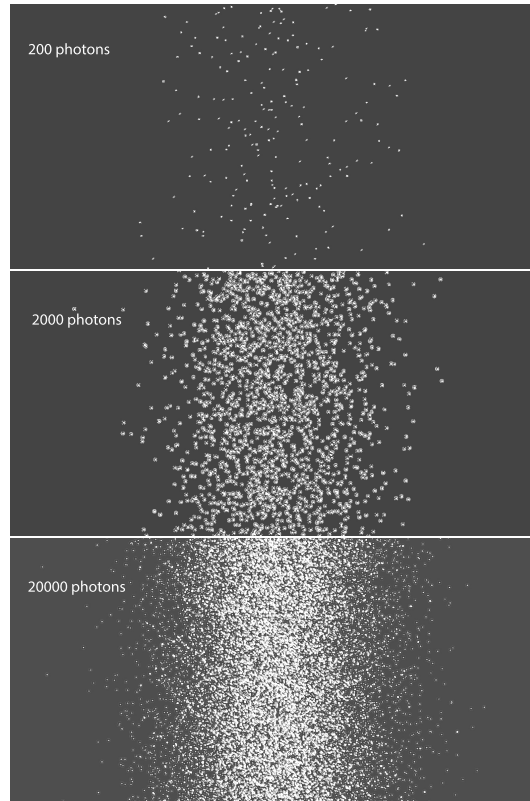


Fig. 4.6: Simulation of the measurement of single photons in a single slit experiment (the dot is where each individual photon hits the detector screen). (Top) 200 photons: the pattern appears completely random. (Middle) 2000 photons: the pattern is starting to emerge but is not clear. (Bottom) 20000 photons: one can see the single-slit pattern forming, but sidebands are still not very clear.

Hence, the two ways to perform the calculation yield exactly the same answer. Why bother with the more abstract quantum way? Because the classical approach requires us to have a continuous electromagnetic wave that we work with and it can be as large or as small in amplitude as we would like. The quantum case has a single photons state—this also has a continuous electromagnetic wave associated with it, but there is a minimal amplitude and it can be only measured once. So, if we run the experiment with a single photon source, we do not know where any individual photon will hit, but the classical probability pattern will instead build up over time, one photon after another (see Fig. 4.6). The quantum approach allows us to consider

this extreme case, which the classical approach is unable to describe. It is a small difference, but an important one. It is the difference between quantum light and classical light.

The quantum effects for the single-slit diffraction experiment are small and subtle. We examine the two-slit diffraction example next. Here, the quantum effects are larger and more profound. But it will take us a few sections to fully develop them. We ultimately will end with what we call the quantum mystery—a fully quantum effect that cannot be explained classically, and we do not even have a clear explanation of it quantum-mechanically!

4.2 Two-Slit Diffraction



Fig. 4.7: Thomas Young who first performed the two-slit experiment in the early 1800s. [Portrait from Henry Perronet Briggs licensed under the Creative Commons Attribution-Share Alike 4.0 International license. Converted to look like a stencil by the authors.]

Newton developed a theory of light during his career which culminated in the publication of his treatise on optics. He felt that light was a particle, because it cast sharp shadows. For centuries, this idea held true, until a British physicist, Thomas Young, illustrated that light did have wave properties by performing the two-slit experiment (see Fig. 4.7). This raised significant controversy and over the decade

after Young's first experiments, Fresnel developed a wave theory of light that was vehemently opposed by Poisson. Poisson came up with a prediction—if one shined light onto a circular mask, then the particle theory says the shadow is completely dark, but the wave theory says there will be a bright spot in the center of the shadow. Arago performed the experiment and saw the bright spot! The wave theory of light was here to stay.

An example of what the two-slit experiment looks like for different colors of light is given in Fig. 4.8. If you compare these images to the single-slit experimental ones in Fig. 4.2, we see that we still have the main central band and the sidebands, but now all of the bands have many new oscillations in them which arise from the presence of the two slits.

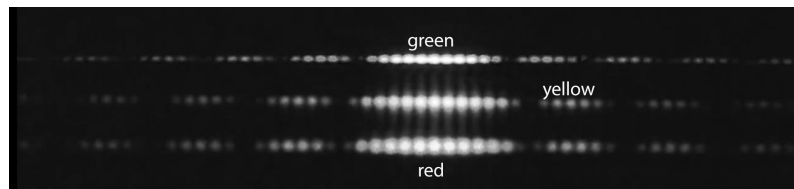


Fig. 4.8: Two-slit experiment using three different colors of light—green (top), yellow (middle) and red (bottom). One can clearly see that the mid-band interference pattern (under the green label and above the red label) and the dimmer subband patterns to the right and the left. In the two-slit experiment, we focus primarily on the central pattern of the single-slit diffraction. Note how each central region is now covered with additional structure of bright and dark features within it. These images were taken from a video on single slit diffraction made by the technical services group of the Massachusetts Institute of Technology [used with permission © Massachusetts Institute of Technology]

In fact, we can show the same phenomenon using a theoretical prediction of the single and two-slit patterns, using the same methods described in Sec. 4.1. When we do this, we see that the single slit pattern, shown in black in Fig. 4.9 is the envelope of the two-slit pattern, shown in gray. In fact, when we discuss the relationship between the two, we usually focus on the bright central region, which now has many nodes in it, and we ignore the side bands of the single-slit diffraction. But, of course, both patterns have wave characteristics to them—they are just much stronger for the two-slit experiment.

Now we now know that light has dual natures of particles and waves, and they are illustrated well with the two-slit experiment, so let us analyze it in detail. Here, we have two narrow slits with a much larger spacing between the slits. Our event is a photon leaves the source, goes through the slits, and is detected somewhere on the screen. If we have slits that are so narrow that we can approximate them with a single path through each of them, then we have two alternative ways for light to go through. If the light we see on the screen for just one slit is 1% of the light that

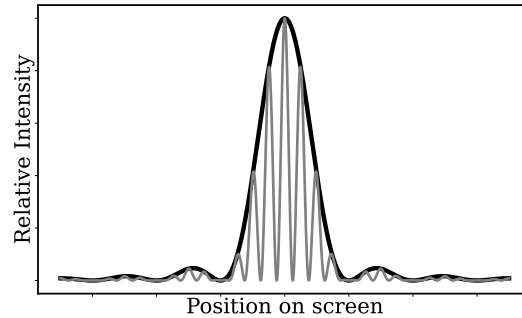


Fig. 4.9: Comparison of the single-slit diffraction pattern (solid black) and the two-slit pattern (gray). One can see the single-slit pattern is the envelope for the two-slit pattern and that the two-slit pattern has a number of nodes inside the bright central peak.

travels straight from the source to the screen, then we will take the length of the arrow to be 0.1, because $(0.1)^2 = 0.01$. So, we have two alternative ways to get from the source to the detector (screen), and hence we have two arrows to determine. If we look at the point on the screen that lines up with the midpoint between the two slits, one can immediately see that the path lengths for the two arrows are the same and hence they must both be in the same direction, so we obtain a total amplitude of 0.2, yielding 4% brightness. As we move up or down away from this midpoint, the two arrows will have different orientations, eventually reaching the point where they are oriented oppositely. Here, when we add them they cancel, and we will see a dark spot. This procedure continues and repeats as we go further out.

It may seem odd to you that before we allowed the system to have interference, we might have predicted if 1% comes through each slit, the brightest peak would be 2%, but our calculation gave 4%. This is because the pattern has changed from a flat curve to one that oscillates. On average, the brightness is about the same, but the pattern changes due to the interference. Furthermore, we do expect that far enough away from the center, the light should fade too. To see this phenomenon, we need to use more alternate ways for the event to occur through each slit. You will investigate this in Prob. 4.3.

The key point of the two-slit experiment is that we will see this interference pattern develop, even if we perform the experiment one photon at a time. The question of how a single photon can do this is a philosophical one, and we have not been able to adequately answer it. Richard Feynman is famous for saying “Nobody understands quantum mechanics.” What he meant by this is no one has figured out how to explain in words how the photon behaves this way.

In Fig. 4.10, we illustrate in a different way the Feynman way to do the calculations. In panel (a), we have the upper slit open, and there are two alternative ways to go from the source to the slit, yielding a large probability at the center, which decays as we go farther away. Panel (b) shows a similar result, and then in (c) we have

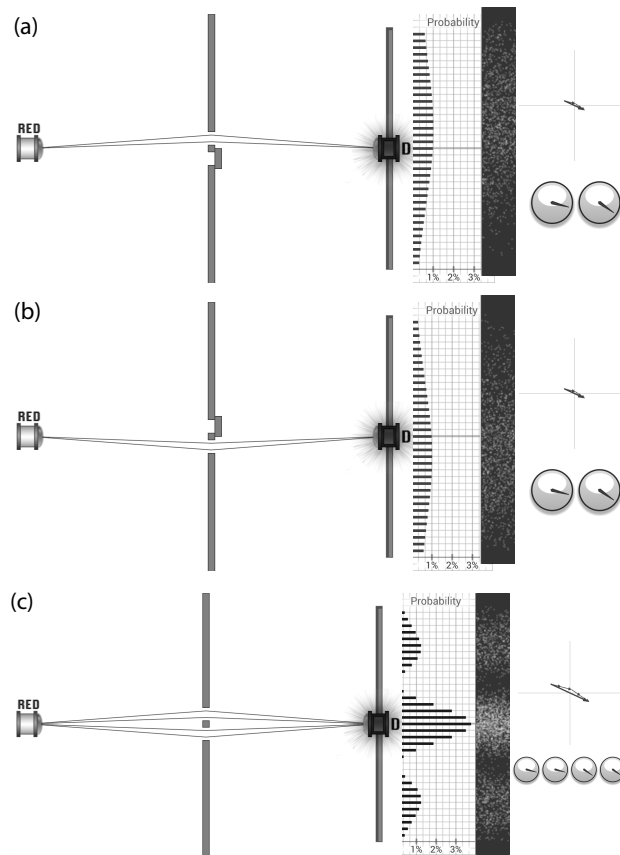


Fig. 4.10: Further illustration of the Feynman approach to describing the two slit experiment. (a) Upper slit open. Here there are two alternative ways to get from the source to the screen. The clocks show the directions of each arrow, and we see they point nearly in the same direction, so we get a large probability. They decay far from the center because the two arrows become anti-aligned. The right image shows how the pattern builds up one photon at a time. (b) Similar to (a), but now with the bottom slit open. (c) Both slits open. Now we have four alternative ways and they can interfere, yielding a more complex probability pattern.

four alternative ways, which interfere giving the more complicated pattern. When they point in the same direction, we have the maximal probability, and when they anti-align, we have a node. All results in between the two will also be seen—the direction of each arrow is found from the time the photon takes to travel from the source to the screen according to the particular alternative way. The small dots show individual photon hits, as the pattern develops one at a time and it is correlated well with the probability pattern.

In addition to the graphical way to describe the system, we want to use Dirac notation and to think of the experiment as state preparation followed by measurement. This also allows us to understand the difference between an abstract quantum state, and a concrete wavefunction, which can be used to find the probability distribution for where the photons will hit the screen at a specific position.

To use Dirac notation, we have to break up our description into the pieces of the experiment. We start with a single photon in the beam heading to the screen with the two slits. Here, our quantum state is $|\text{beam}\rangle$. Now, when we go through the slits, we create a superposition of the two alternative ways to go through (through slit A or through slit B). We do not know whether it went through one or the other, we do not know that it went through both, we only know that it went through in superposition (or we can also say it went through in an indeterminate way). The state then becomes $|\psi\rangle = 0.1(|\psi_A\rangle + |\psi_B\rangle)$, where the amplitude of each arrow is 0.1 in size after passing through the slits. Finally, we measure at a position x on the screen. Measurement requires us to project onto the state corresponding to what we measure, so we multiply by the ket-bra “butterfly” $|x\rangle\langle x|$, which projects us onto the position near x . This gives

$$0.1(\langle x|\psi_A\rangle + \langle x|\psi_B\rangle)|x\rangle = \langle x|\psi\rangle|x\rangle \quad (4.2)$$

and the probability for this event to occur is the modulus squared of this state. Using $\langle x|x\rangle = 1$ (which is true if we localize the position to a small, but not infinitesimal region in space), we find

$$\begin{aligned} P(x) &= |\langle x|\psi\rangle|^2 \\ &= 0.01(|\langle x|\psi_A\rangle|^2 + |\langle x|\psi_B\rangle|^2 + \langle\psi_A|x\rangle\langle x|\psi_B\rangle + \langle\psi_B|x\rangle\langle x|\psi_A\rangle) \\ &= 0.01(|\psi_A(x)|^2 + |\psi_B(x)|^2 + 2\text{Re}(\psi_A^*(x)\psi_B(x))). \end{aligned} \quad (4.3)$$

Here, we use the standard notation for a wavefunction, given by $\langle x|\psi\rangle = \psi(x)$. As x changes, the probability changes, because we need to determine the two arrows for each x point, and they change because the time the photon traveled changes for each position.

Hence, we can think of the quantum state as the set of rules we will use to evaluate what the photon does, while the wavefunction corresponds to applying those rules concretely at a specific position x to get a final arrow, and the probability distribution is the modulus squared of the final arrow at each position. It is useful to keep this notion clear in your head about the difference between a state and a wavefunction. The graphic in Fig. 4.11 illustrates these concepts.

We are only half-way through our story now. There is more exciting stuff to come, but before we get to that, we need to discuss another property of light, which is polarization.

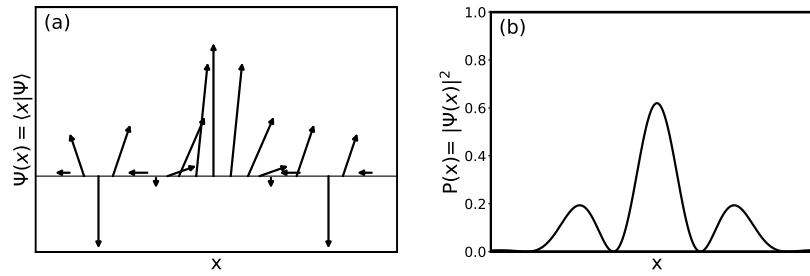


Fig. 4.11: (a) Wavefunction and (b) probability distribution for the two-slit experiment as functions of the position x . As we mentioned, the quantum state is the set of rules we use to evaluate the probability amplitude at any point. The wavefunction (a) evaluates at a specific position (where we plot the final arrow) and the probability distribution (b) is the modulus squared of each final arrow at each position.

4.3 Polarization of Light

You should already know that light is an oscillating electromagnetic field. It is a transverse wave, so it moves in some direction, and the electric and magnetic fields lie in the plane perpendicular to the direction of the motion. The fields themselves are also perpendicular, so the electric field is in some direction, and the magnetic field is perpendicular to it as well. The direction that the electric field points in is called the polarization of the light. We will assume in our work here that this direction does not change when the photon is moving through a vacuum. This case corresponds to linearly polarized light. You can think of this as similar to the spin degree of freedom we discussed with atoms in Chap. 2. This is only a loose analogy, as the two concepts have clear differences.

There is an important piece of equipment called a polarizer that we can use to manipulate and control the polarization of light. A polarizer is like a picket fence, and the photon has a polarization direction it carries with it, like a little rod (see Fig. 4.12). If the rod points in the same direction as the picket fence, then it goes through. If it is perpendicular to the pickets it hits them and is stopped. So, a polarizer acts like a filter that filters out all light that is not polarized in the same direction. The light that is in the wrong polarization is absorbed. How does this work for photons? Well, for a given polarization of the photon and a given orientation of the polarizer, there will be a probability for the photon to pass through. If it does, it has the polarization of the polarizer. If it does not get through, it was absorbed. An example is shown in the right panel of Fig. 4.12. One can see the unpolarized light has the directors pointing in any possible direction, but once we pass through the horizontal polarizer, they are pointing horizontally. Then if we try to pass them through a vertical polarizer, they are all absorbed.

Note that light oscillates, so, if the electric field is pointing up at some point of time, later, we can find it pointing down. This means that light that is polarized up

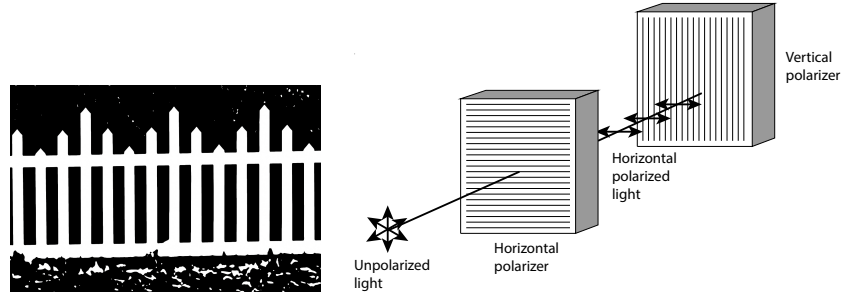


Fig. 4.12: (left) Picket fence and (right) polarization graphic. The picket fence is an analogy for a polarizer letting polarization of only one direction through. The graphic on the left shows that we can think of unpolarized light as having any polarization. Once it gets through a horizontal polarizer it will be horizontally polarized. Then trying to pass it through a vertical polarizer will fail and all the light will be absorbed. [Picket fence from image from ScottWebb on pexels.com and converted to look like a stencil by the authors.]

goes through the same polarizer as light that is polarized down, so the direction of the polarization is neither up nor down but instead is something we call “up-down.” We use a line segment with no arrows to describe it, which we call a director. What direction does it point in? Two directions! Both up and down, or both right and left, or both in and out.

Let us now delve deeper into the concept of polarization. We will use the Dirac notation to describe it with the basis of states given by $|V\rangle$ for vertical polarization and $|H\rangle$ for horizontal polarization. Then a state polarized at an angle θ is given by $\cos \theta |V\rangle + \sin \theta |H\rangle$. The states $|V\rangle$ and $|H\rangle$ are orthonormal states, so $\langle H|V\rangle = 0$. An unpolarized state will be denoted by $|\text{un}\rangle$, which is a simplified way to denote a state which is an equal probabilistic mixture of $|V\rangle$ and $|H\rangle$. It can equally well include a probabilistic mixture of states in all possible directions as well. To fully describe this state we need to discuss density matrices, which we will not be doing here. But the concept of a uniform mixture of states in any direction is a simple picture to use and this will work fine for our purposes.

Now, we wish to use our Dirac notation to discuss state preparation of light in experiments with polarizers. As shown in Fig. 4.13. Starting in panel (b), we have a photon state leaving the source that is in the $|\text{un}\rangle$ state. Now, when it passes through the polarizer, it will emerge in the $|V\rangle$ state, or it will be absorbed. All of the light that emerges is in the $|V\rangle$ state, so we say we have a transition $|\text{un}\rangle \rightarrow |V\rangle$ for the photons that are not absorbed. Then it proceeds further, hits the screen and is observed.

That was a simple experiment. Now, we want to examine something similar to the three analyzer experiment we did with the Stern-Gerlach analyzer loops with closable gates in Prob. 2.8. The experiment is shown in Fig. 4.14. The first case, in panel (a) is the analogue of Fig. 2.24. The first polarizer takes us from $|\text{un}\rangle \rightarrow |V\rangle$. Passing through a polarizer when we are in a definite (or pure) state is calculated by applying

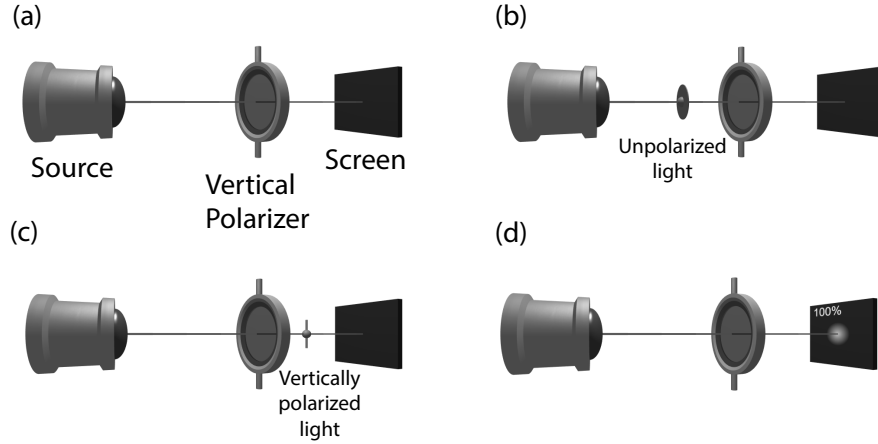


Fig. 4.13: Passage of light from an unpolarized source through a polarizer and then hitting a screen. (a) The apparatus used: (i) source; (ii) a vertical oriented polarizer (indicated by the little rods); and (iii) a screen that the light hits. (b) Initially the light comes out in an unpolarized state where we do not know what direction it is in. (c) After passing through the vertical polarizer, its direction is up-down. Then it hits the screen and all the light that passed through the polarizer hits the screen.

a projection operator corresponding to the state that emerges through the polarizer. So passing through the second polarizer yields $|\text{un}\rangle \rightarrow |V\rangle \rightarrow |V\rangle \langle V|V\rangle = |V\rangle$. So 100% of the light that passed through the first polarizer also passes through the second one. The second case is like Fig. 2.23. Here the process is $|\text{un}\rangle \rightarrow |V\rangle \rightarrow |H\rangle \langle H|V\rangle = 0$. Hence, no photons make it to the screen. The third case is the analogue of Fig. 2.26. Here, we have the following process:

$$|\text{un}\rangle \rightarrow |V\rangle \rightarrow |45^\circ\rangle \langle 45^\circ|V\rangle = \frac{1}{\sqrt{2}} |45^\circ\rangle \rightarrow \frac{1}{\sqrt{2}} |H\rangle \langle H|45^\circ\rangle = \frac{1}{2} |H\rangle, \quad (4.4)$$

and hence, 25% of the light ($0.5^2 = 0.25$) that made it through the V polarizer will hit the screen. Here, we used the fact that $|45^\circ\rangle = \frac{1}{\sqrt{2}} |V\rangle + \frac{1}{\sqrt{2}} |H\rangle$. Note how, just like we discovered with spins, photons behave similarly—they remember only the last axis they were measured on—for spins this is the orientation of the analyzer, for photons, the orientation of the polarizer.

This concludes our brief discussion of polarization and how to work with it using Dirac notation and our quantum rules.

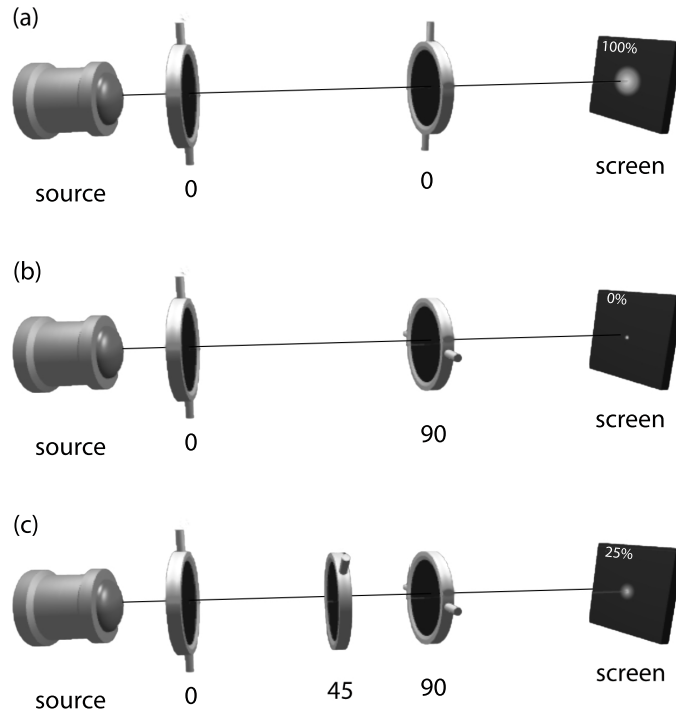


Fig. 4.14: Three experiments with polarizers: (a) vertical and vertical polarizer; (b) vertical and horizontal polarizer; and (c) vertical, diagonal, and horizontal polarizer.

4.4 The Quantum Mystery

We now have all of the machinery we need to discuss the quantum mystery of the two-slit experiment. This shows all of the bizarre features of the quantum world, and many other weird examples can be related back to this experiment and how it behaves. We call it a mystery, because we do not have any simple way to use words to describe how it works. Our mathematical description, however, is ironclad and clear—there is no mystery at all. But, we like to use words to describe how quanta can behave this way, and the situation we will unveil for you here remains frustrating in this respect.

We start by recapping the behavior of the two-slit experiment when we have one alternative way that we can pass through each slit. This corresponds to panel (c) of Fig. 4.10, but modified to have one alternative way through each slit. Let us describe the behavior using our formalism. The light starts in a beam coming from the source, which we call $|\text{beam}\rangle$. If we assume that 1% of the photons incident on either slit will pass through, the act of passing through a slit requires us to multiply the state by 0.1.

So, after passing through the slit we have $|\text{beam}\rangle \rightarrow |\psi\rangle = 0.1(|\psi_A\rangle + |\psi_B\rangle)$, with the two slit states being orthonormal states; note that this state $|\psi\rangle$ is not normalized, but this is okay for how we address this particular problem which is mirroring the Feynman approach. This state is called a superposition state, where the state of the photon is indeterminate as it travels to the screen. Now, the photon moves to the screen and will be measured at some position. Our theory answers the question of what is the probability to measure the photon at position x . The answer is $|\langle x|\psi\rangle|^2$. As mentioned before, the quantum state $|\psi\rangle$ encodes the quantum rules, while the wavefunction is a concrete evaluation of those rules at a specific position.

Now, we know the rules to follow. Each alternative way has an arrow associated with it, of length 0.1 and pointing in the direction determined by the time it took to move from the source to the screen and the color of the light. We simply think of $0.1 \langle x|\psi_{A,B}\rangle$ as those two arrows evaluated at position x . When we do this, and look over all x values, we see the interference pattern, shown in Figs. 4.8 and 4.11.

As we mentioned at the start of this section, we do not know how to explain how this is done. Our standard description is that if we do this experiment with single photons, so the pattern develops over time one-by-one, then somehow, the photon knows about both slits. How does it know this? Does it travel through one or the other, or both? We are careful to say we just do not know—its state is indeterminate. But, this leaves us with an uneasy feeling. Can't we find a way to watch them move through each slit and find out how a photon magically knows how to build up such a pattern?

The first thought is to just use detectors to watch as they move past each slit. But, we saw in Chap. 1 that photons can be measured once and only once, because they are destroyed in most photon detectors, so no nondestructive detectors are available. Does this mean we are stuck? No! We can use the other property of photons, their polarization, to allow us to “tag” each photon as a means to tell which slit it went through. We follow recent discussions about how to do this in our treatment here [3, 2].

We start with Fig. 4.15, which shows the set-up of the experiment. We start with a nearly monochromatic light source and we use a diagonal polarizer to create the initial state, which is given by $|\psi\rangle = |\text{beam}\rangle \otimes |D\rangle = |\text{beam}\rangle \otimes \frac{1}{\sqrt{2}}(|V\rangle + |H\rangle)$. This is a product state superposition, because the state factorizes across the tensor product sign, as the rightmost expression clearly shows. Next, the photon passes through the slit. We note that most of the photons are absorbed either by the screen or by the polarizers. But, the ones that do get through do so in an indeterminate state—they either are in $0.1 |\psi_A\rangle \otimes |V\rangle$ or in $0.1 |\psi_B\rangle \otimes |H\rangle$, with equal amplitude, so the state becomes

$$|\psi\rangle = \frac{0.1}{\sqrt{2}} (|\psi_A\rangle \otimes |V\rangle + |\psi_B\rangle \otimes |H\rangle). \quad (4.5)$$

The 0.1 comes from the fact that only 1% of the photons incident on a single slit make it through, and the $\sqrt{2}$ in the denominator arises from the fact that when we pass through a polarizer, we must multiply the polarization part of the quantum state by the corresponding projector (here, $|V\rangle\langle V|$ or $|H\rangle\langle H|$) corresponding to the measurement that happens at the polarizer and we have that $\langle D|V\rangle = \frac{1}{\sqrt{2}}$ and

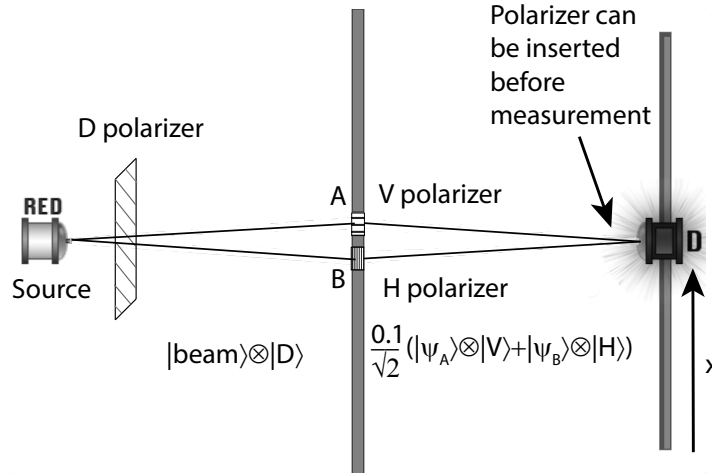


Fig. 4.15: Schematic of a two-slit experiment with polarizers. One creates a polarized source with the initial polarization given by $|D\rangle = \frac{1}{\sqrt{2}} (|V\rangle + |H\rangle)$. Then, the light that makes it through the slits will create an entangled superposition, given by $\frac{0.1}{\sqrt{2}} (|\psi_A\rangle \otimes |V\rangle + |\psi_B\rangle \otimes |H\rangle)$, which is an entangled state. When measured, the entangled state will show no interference. So, by tagging the photons to determine which-way information, the interference pattern is destroyed.

$\langle D|H\rangle = \frac{1}{\sqrt{2}}$. If we look at the norm of this state, it is 0.1, which can be understood as 1% through each slit but, half of the photons that could have gotten through the slit are absorbed by the polarizers because they have the wrong polarization. It is important to note that passing through the polarizers is a measurement because it is irreversible—we cannot construct the polarization of the photon before it went through the polarizer from information we have after it passed through.

Now this state is in what we call an entangled state, because it cannot be factorized across the tensor-product sign (try it to be sure). The behavior in an entangled state is different from what it is in a product-state superposition. What we will find when we measure at the screen is that the interference all goes away and we have a flat probability distribution. To see this, we recall that measuring at the screen implies we project onto the state $|x\rangle$, which acts only on the spatial ψ part of the quantum state. It forms the wavefunction for each component. Now, if we assume the measurement at x is over a small interval Δx , which is the size of the detector, then we can assume that the x state of the detector is a normalized state with $\langle x|x\rangle = 1$. Hence, the measurement process takes us to the state

$$|\psi_{\text{meas}}\rangle = \frac{0.1}{\sqrt{2}} (\psi_A(x) |x\rangle \otimes |V\rangle + \psi_B(x) |x\rangle \otimes |H\rangle) \quad (4.6)$$

after projecting onto the detector state at x , given by $|x\rangle \langle x|$. Now, we take the modulus squared of the state (or the norm of the state) to determine the probability to be measured at x . This is

$$\begin{aligned} P(x) &= 0.005(|\psi_A(x)|^2 \langle x|x\rangle \langle V|V\rangle + \psi_A^*(x)\psi_B(x) \langle x|x\rangle \langle V|H\rangle \\ &\quad + \psi_B^*(x)\psi_A(x) \langle x|x\rangle \langle H|V\rangle + |\psi_B(x)|^2 \langle x|x\rangle \langle H|H\rangle) \\ &= 0.005(|\psi_A(x)|^2 + |\psi_B(x)|^2) = 0.01, \end{aligned} \quad (4.7)$$

because $\langle V|H\rangle = \langle H|V\rangle = 0$. Note that this is the sum of the length-squared of each of the Feynman arrows, so we see no interference effects.

Hence, when we try to look to see how the photon interferes one particle at a time, we find that it does not. If we know which slit the photon goes through, then it actually goes through just one slit, and there is no interference pattern. If we do not know which slit the photon goes through, we get the interference pattern. This principle is called complementarity—when we have which-way information, there is no interference, but when we do not have which-way information, the system can interfere.

Nature is stubborn and will not reveal its secrets. In spite of running a clever experiment that allows us to know which slit the light goes through, without disturbing the photon in the process, we still are unable to learn how a single photon interferes with itself.

Note that we have not fully explained how we know what slit the photon went through. Indeed, in the measurement we just discussed, we actually do not know which slit each photon went through, because we did not attempt to measure this. Just the sheer fact that we could measure which slit the photon went through is enough for us to lose the interference pattern. What if we really want to know which slit it went through? This can be done by adding a polarizer just before measuring. Now, we project with the state $|x\rangle \otimes |V\rangle \langle x| \otimes \langle V|$ (and a similar formula if we want to measure the horizontal polarizer). If we use a vertical polarizer, the state after measuring is $\frac{0.01}{\sqrt{2}}\psi_A(x)|x\rangle \otimes |V\rangle$ and the probability becomes $0.005|\psi_A(x)|^2$, which is clearly the probability to go through slit A. If we used the horizontal polarizer, then we would find a similar result with $A \rightarrow B$ and $V \rightarrow H$.

This type of experiment, where one can choose just before the final measurement what set-up one will actually measure is called a delayed-choice experiment, and it was devised by the famous physicist John Wheeler (who was Feynman's Ph.D. adviser). You can see an image of John Wheeler in Fig. 4.16. John Wheeler was a successful and influential physicist in his own right. He co-wrote a wonderful set of relativity books that explain the ideas without requiring a significant math requirement [5, 6] and a famous advanced textbook on general relativity [4]. Our interest, however, is his idea of the delayed choice experiment [7], where we decide just before measuring, precisely what we will measure. We have seen two choices so far—using a vertical or a horizontal polarizer. What happens if we use a diagonal polarizer? Then the projection operator we multiply by is



Fig. 4.16: John Wheeler in the 1950s. Wheeler was an influential physicist of his day. He worked primarily in gravity, but had a strong interest in quantum mechanics and had a number of innovative ideas, including the idea for a delayed-choice experiment. He even wanted to perform the experiment using the two images of a distant quasar that appear when the system is gravitationally lensed by a black hole in between the quasar and us. [Wheeler image from Ulli Steltzer, courtesy of AIP Emilio Segrè Visual Archives and converted to look like a stencil by the authors.]

$$|x\rangle \otimes |D\rangle \langle x| \otimes \langle D| = \frac{1}{2} |x\rangle \otimes (|V\rangle + |H\rangle) \langle x| \otimes (\langle V| + \langle H|). \quad (4.8)$$

Applying this to our state gives us the state

$$|\psi\rangle = 0.05(\psi_A(x) + \psi_B(x)) |x\rangle \otimes |D\rangle, \quad (4.9)$$

which is a product state superposition. Now, when we compute the probability, we find it is equal to

$$P(x) = 0.0025 |\psi_A(x) + \psi_B(x)|^2, \quad (4.10)$$

which, indeed, is the situation where we add the two Feynman arrows, determine the final arrow, and then square to obtain the probability—the situation where we see interference.

We have a cautionary note about this discussion as it is often described as showing something called retrocausality—where the quantum system must go back in time to determine whether to interfere or not. The idea behind this is that the interference is determined back at the screen with the two slits, so one must go back in time to restore interference. Our analysis clearly shows this discussion is complete nonsense. Instead, for any quantum system, we simply prepare the state according to what the experiment does and we measure the state at the end after all preparations have been completed. Then whether we have interference or not simply arises from whether we have a product state superposition (which has interference) or an entangled superposition (which does not have interference).

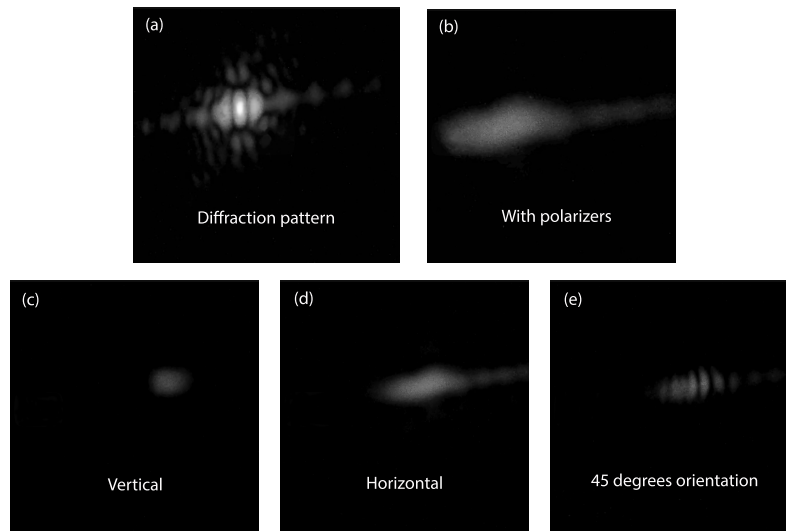


Fig. 4.17: Example of a delayed choice experiment using polarizers by Paul Nord of Valparaiso University. His experiment uses polarizers on two sides of a thin metal wire, but the math and the discussion is identical to the two-slit experiment we discussed. We have (a) the diffraction pattern with no polarizers; (b) the diffraction pattern with polarizers but no measurement of polarization; (c) a V polarizer before measurement; (d) and H polarizer before measurement; and (e) a diagonal polarizer before measurement. One can clearly see all of the behavior we discussed about this experiment in the images and now you know how to explain them all yourself. [Image courtesy of Paul Nord.]

This effect is real, as you can see by searching for delayed choice two-slit experiment and watching correct videos on the topic (if they discuss retrocausality as being true, you know they are wrong). Here, we show a beautiful demonstration of the effect by Paul Nord, a professor at Valparaiso University. In Fig. 4.17, one can see the diffraction pattern with no polarizers on the screen with the slits (a), with polarizers (b), where the interference pattern disappears. Then using a V polarizer (c), an H polarizer (d), or a diagonal polarizer (e), we see the pattern from one slit, or the other, or both in a situation where the interference is restored. One can really do these kinds of experiments! As a technical note, the Paul Nord demo is organized a little bit differently using diffraction off a thin metal wire and with polarizers to the left or the right of the wire, but the math is identical and so is the explanation.

After all of this work, we clearly see the formalism can easily describe when we have and when we do not have interference. The mystery we have is how does the photon know to behave this way and make the right pattern one photon at a time? It is the mechanism we do not have a clear description for, even though we have a perfect mathematical formalism for the experiment.

4.5 Mach-Zehnder Interferometer



Fig. 4.18: (Left) Ludwig Mach and (Right) Ludwig Zehnder, both images from the 1890s, when they worked on developing the Mach-Zehnder interferometer. [Mach image: Photographer Wilhelm Felchner, public domain. Zehnder image: Public domain. Both images modified to look like stencils by the authors.]

We end this chapter with a discussion about the Mach-Zehnder interferometer. It has a strong analogue with the Stern-Gerlach analyzer loop, in that it puts the photon into an indeterminate state as a superposition, separates the states so one can manipulate different parts of the superposition separately, and then rejoins them back together. Just like the analyzer loop, if we do nothing to the photon inside the interferometer (and it is tuned properly) then it does nothing. But, if we do something, it is very sensitive in showing us the implications of what we have done.

The interferometer was created by Ludwig Mach and Ludwig Zehnder, with Mach improving upon a design by Zehnder. Mach was the son of the famous Physicist and Philosopher Ernst Mach and supposedly led a somewhat troubled life, most likely due to struggles with mental health. Zehnder was a student of Röntgen, and was the person who actually took the first x-ray image of Röntgen's wife's hand with her wedding ring on it.

We start by describing the interferometer, which is shown in Fig. 4.19. It requires some new rules for light to treat them in the Feynman fashion. We start with nearly monochromatic light from the light source. This light is incident on a 50-50 beam splitter. The beam splitter is like partially reflecting sunglasses, where half of the light is reflected and half is transmitted. Note that when we reflect from air, as is done in this interferometer, we add six hours to the clock rotation. The transmitted

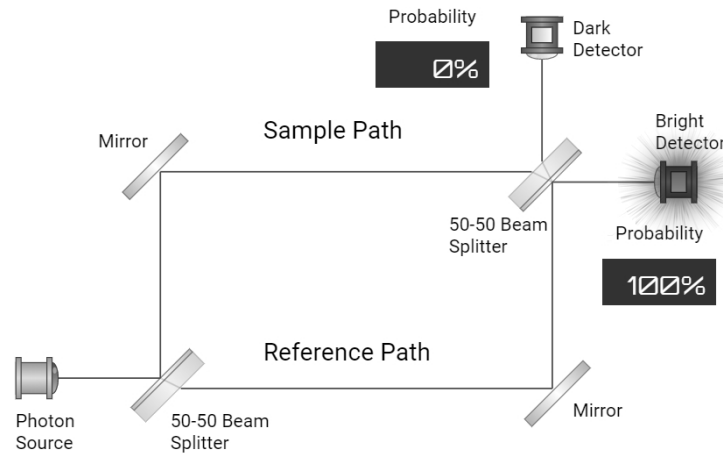


Fig. 4.19: Schematic of the Mach-Zehnder interferometer. It starts with a nearly monochromatic light source, which sends a light beam into one of the input ports of the beam splitter, which equally separates the light into two alternative paths—transmission on the reference path and reflection on the sample path. Then after reflecting in a mirror on each path, the two paths converge on a second beam splitter with output going to two detectors—one called the dark detector and the other the bright. Note that while the first beam splitter reflects from air, the second reflects from glass. If the path lengths of the two paths are an integer multiple of the wavelength of the light, then all of the light goes to the bright detector and none to the dark detector, which is the case shown in the figure. This is the origin of the two names.

light goes on the reference path and the reflected light travels on the sample path. The two paths meet at the second 50-50 beam splitter where they can be reflected or transmitted and end up at the bright or dark detector. Note that the orientation of the beam splitter is flipped here, so that the reflection comes from glass and thereby we add no hours to the clock upon reflection. The path length on the sample path is adjusted by moving the mirror on the sample path inward or outward by a small amount (this is typically done with piezoelectrics).

Let us carefully analyze the interferometer using the Feynman approach. We consider the two alternative ways to go to the bright detector. For the reference path way, we start with the clock hand at 12 o'clock. It rotates as the photon moves from the source to the beam splitter. When we go through the first beam splitter, we transmit, so we shrink the arrow by a factor of $\frac{1}{\sqrt{2}}$. It then moves along the reference path, with the arrow rotating until we hit the mirror. Here, the reflection adds 6 hours (or a phase of π) to the clock. Then it rotates as we move to the second beam splitter. Here, we reflect off air (so add six hours or a phase of π) and we shrink by $\frac{1}{\sqrt{2}}$. Then we rotate until we hit the detector. The total phase is

$$\begin{aligned}\phi_{\text{tot}}^{\text{reference-bright}} &= \phi_{\text{source-splitter}} + \phi_{\text{splitter-mirror}}^{\text{reference}} + \phi_{\text{mirror-splitter}}^{\text{reference}} + \pi \\ &\quad + \phi_{\text{splitter-bright detector}} + \pi.\end{aligned}\tag{4.11}$$

For the sample path way, we have the clock hand at 12 o'clock. It rotates as the photon moves from the source to the beam splitter. When we go through the first beam splitter, we reflect, so we shrink the arrow by a factor of $\frac{1}{\sqrt{2}}$ and add six hours (or a phase of π). It then moves along the sample path, with the arrow rotating until we hit the mirror. Here, the reflection adds 6 hours (or a phase of π) to the clock. Then it rotates as we move to the second beam splitter. Here, we transmit (so add six hours or a phase of π) and we shrink by $\frac{1}{\sqrt{2}}$. Then we rotate until we hit the detector. The total phase is

$$\begin{aligned}\phi_{\text{tot}}^{\text{sample-bright}} &= \phi_{\text{source-splitter}} + \pi + \phi_{\text{splitter-mirror}}^{\text{sample}} + \phi_{\text{mirror-splitter}}^{\text{sample}} + \pi \\ &\quad + \phi_{\text{splitter-bright detector}}.\end{aligned}\tag{4.12}$$

Obviously the phase accumulated from source to beam splitter and from beam splitter to detector are the same for both ways. We are told the phase accumulated on the sample and reference paths, including the reflection off the mirror, are also the same for both ways. Hence, one can see that $\phi_{\text{tot}}^{\text{reference-bright}} = \phi_{\text{tot}}^{\text{sample-bright}}$. This means the two arrows, each of length 0.5, are pointing in the same direction. Adding them together, we see they become a final arrow of length one, so 100% of the photons go to the bright detector.

A similar analysis for the dark detector, shows that the phase for each alternative way differs by exactly π , so the arrows are pointed opposite each other and will cancel, yielding a final arrow of length 0, so no photons go to the bright detector. Be sure to carefully go through this analysis yourself.

Now, if we move the mirror on the sample path, we can add or subtract from the phase along that path. This will rotate the arrows for each alternative way that is on the sample path by the same amount. The reference path remains unchanged. We will cycle the probabilities from 100% to 0% and then back to 100% on the bright detector and from 0% to 100% to 0% on the dark detector. Note that for each additional phase, the total probability remains 1, which is discussed in Prob. 4.8.

What is the interferometer used for? Well it is sensitive to distances on the order of about 0.01 the wavelength of light, so a few nanometers, and it can be employed to visualize small changes in the phase of the light. One can use it to visualize smoke, for example, since the smoke adds phase according to its density to the light. It's main application though is to compare distances with high accuracy. This is what it does in the LIGO observatory, which measures gravitational waves and will be discussed in Chap. 6.

Now, let us repeat the analysis using the Dirac notation. We let $|\psi_{\text{sample}}\rangle$ denote the state along the sample path and $|\psi_{\text{reference}}\rangle$ denote the state along the reference path. This state will accumulate the phase along the paths as the photon moves through the interferometer; phases from reflections are accumulated outside the state. Then, the state preparation for going to the bright detector is

$$\begin{aligned}
|\text{beam}\rangle &\rightarrow \frac{1}{\sqrt{2}}(-|\psi_{\text{sample}}\rangle + |\psi_{\text{reference}}\rangle) \rightarrow \frac{1}{\sqrt{2}}(|\psi_{\text{sample}}\rangle - |\psi_{\text{reference}}\rangle) \\
&\rightarrow \frac{1}{2}(|\psi_{\text{sample}}\rangle + |\psi_{\text{reference}}\rangle).
\end{aligned} \tag{4.13}$$

Here we start with the beam heading to the beam splitter, then we have the state right after the beam splitter, then after reflecting off the mirrors and finally after the second beam splitter (and on the bright-detector path). Then, taking the overlap with $\langle \text{bright} |$ gives us the final probability arrow. We square this to determine the probability—recall that this means evaluate our general quantum rules at the location of the bright detector. This yields

$$P_{\text{bright}} = \frac{1}{4} |e^{i\phi_{\text{sample}}} + e^{i\phi_{\text{reference}}}|^2. \tag{4.14}$$

A similar analysis for the dark detector yields

$$P_{\text{dark}} = \frac{1}{4} |e^{i\phi_{\text{sample}}} - e^{i\phi_{\text{reference}}}|^2. \tag{4.15}$$

You should be becoming familiar with how we go back and forth between the Feynman approach and the Dirac notation method, and how one generates final concrete numbers for each approach.

We would also like to discuss what happens if we use polarized light and add polarizers on the two detectors. As we did with the two-slit experiment, we will send in diagonally polarized light from the source. Then, we will use a V polarizer on the sample path and an H polarizer on the reference path, both placed after the mirrors. Now, the state preparation for the two alternative ways that we reach the bright detector becomes

$$\begin{aligned}
|\text{beam}\rangle \otimes \frac{1}{\sqrt{2}}(|V\rangle + |H\rangle) &\rightarrow \frac{1}{2}(-|\psi_{\text{sample}}\rangle + |\psi_{\text{reference}}\rangle) \otimes (|V\rangle + |H\rangle) \\
&\rightarrow \frac{1}{2}(|\psi_{\text{sample}}\rangle \otimes |V\rangle - |\psi_{\text{reference}}\rangle \otimes |H\rangle) \\
&\rightarrow \frac{1}{2}(|\psi_{\text{sample}}\rangle \otimes |V\rangle + |\psi_{\text{reference}}\rangle \otimes |H\rangle).
\end{aligned} \tag{4.16}$$

This state preparation starts with the state before the first beam splitter, then after the first beam splitter, after the mirrors and polarizers, and finally after the second beam splitter on the bright-detector path.

Now, if we measure without looking at the polarization, we find no interference effects, because $\langle V|H\rangle = 0$ and the total probability is 25%. We obtain the same probability for the dark detector. Recall that half of the photons are absorbed by the polarizers when they are measured by them. Note that the loss of interference arises from the fact that we have which-way information encoded in the polarization and the principle of complementarity.

If we put a V polarizer before measuring at either detector, we obtain only the photons that traveled the sample path. We can do this at either detector. A similar result from the reference path occurs when we use an H polarizer. What is interesting is using a $|D\rangle$ polarizer. Here, we obtain 25% at the bright detector and 0% at the dark detector, because we restore the product-state superposition. Again, we can apply the diagonal polarizers to either detector or both. Hence, one can see that the Mach-Zehnder interferometer has quite similar behavior to the two-slit experiment with polarizers, just for the interferometer, we do not have large patterns we look at, just the counts in each detector.

4.6 Summary

This chapter summarized behavior that has no classical counterpart. While the behavior of the single and two-slit diffraction patterns become the classical pattern after many photons have been observed, the patterns can be built up one photon at a time, indicating that photons, which are in an indeterminate state (and travel in superposition) have the probability pattern encoded in their dynamics, even though we ultimately measure them at only one specific location on the screen.

Our discussion used both the Feynman path integral approach (with arrows for each alternative way added into a final arrow) and a Dirac-notation approach (which worked with states in superposition). Both approaches analyze the experiments in the same way and yield the same answers. You can use either to reason through how quantum systems behave. They are both very powerful techniques for qualitative and quantitative reasoning about quantum experiments. Few people know how to use them effectively, so you can put yourself ahead of your peers once you master how to use them. We will continue to practice with these methods in the next two chapters.

When we try to develop an experiment to measure which path the photons go through in the two-slit experiment, we find that they do not go through both slits, as you might suspect, but they go through one or the other. But, now the probability pattern has changed from an interference pattern to the sum of the single-slit pattern from each slit. Nature does not allow us to easily determine how the single photon knows how to create the classical probability pattern one photon at a time. Of course, the theoretical formalism handles all of this perfectly, because the formalism only determines the probability pattern and cannot answer the question of what any individual photon will do. This behavior is called the quantum mystery, and it underlies a behavior that cannot be described with any classical theory.

We used the polarization of light to tag the photons according to the slit they went through, thereby creating an entangled state. We showed how entanglement destroys interference effects. We even showed a delayed-choice experiment, where the interference can be restored by changing the entangled superposition state into a product-state superposition. This exercise affirms for us that we describe experiments by a state preparation phase followed by a measurement phase. The measurement

is always performed on the final state prepared. At no point in time do we need to resort to retrocausality to describe quantum behavior.

We ended with a discussion of the Mach-Zehnder interferometer, which displayed behavior similar to both the Stern-Gerlach analyzer loop and the two-slit experiment. It was analyzed by the Feynman approach as well. We also showed that adding in polarizers to the experiment enriches the different behaviors we can see with it.

Chapter 4 problems

Problem 4.1 Single-slit diffraction In this problem, we discuss how to calculate the single-slit diffraction intensity as a function of the angle from the center of the slit to the position of the screen. We assume the slit runs from $-\frac{a}{2}$ to $\frac{a}{2}$. We need to sum the Feynman arrows, of the form $e^{i\phi}$ for each point along the wavefront, whose angle is given by the extra phase traveled from the slit to the wavefront. This is given by the phase,

$$\phi = \frac{2\pi(x + \frac{a}{2}) \sin \theta}{\lambda}. \quad (4.17)$$

The complex amplitude of the light is given by

$$A = \int_{-\frac{a}{2}}^{\frac{a}{2}} dx e^{i\phi(x)}. \quad (4.18)$$

Plot the intensity $|A|^2$ as a function of θ for $\frac{a}{\lambda} = 10$ and for $-\frac{\pi}{3} \leq \theta \leq \frac{\pi}{3}$. Explain why the integral does determine the sum of the Feynman arrows.

Problem 4.2 Essay on single-slit diffraction. Explain how the single slit diffraction pattern changes as the slit changes its width. Include also a discussion for a fixed width slit, where we change the color of the light (which is the same as changing the wavelength of light).

Problem 4.3 Computation of the two-slit pattern. We calculate the probability distribution for the two-slit experiment using two arrows in each slit, as shown in Fig. 4.10(c), with two arrows in each slit. Let the angle for the arrow that represents the alternative way of the top path be $\theta + \frac{\alpha}{2} + \frac{\phi}{2}$ and of the second path in the upper slit be $\theta + \frac{\alpha}{2} - \frac{\phi}{2}$. Then we take the angle of the upper path in the lower slit to be $\theta - \frac{\alpha}{2} + \frac{\phi}{2}$ and of the lower path in the lower slit to be $\theta - \frac{\alpha}{2} - \frac{\phi}{2}$, with $-\pi \leq \phi \leq \pi$ and with $\alpha = 3\phi$. Note that your probability will be independent of θ . Plot the probability curve as a function of ϕ .

Problem 4.4 Three-slit experiment. Consider the three-slit experiment as depicted in Fig. 4.20, with one path through each slit. Using the Feynman method with arrows, discuss how the probability pattern will change as one moves from the center of the screen opposite the middle slit, up to the upper edge of the screen.

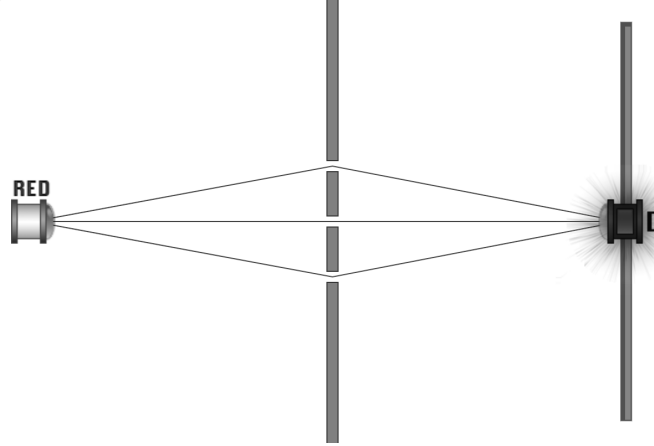


Fig. 4.20: Three-slit experiment with one alternative way through each slit.

Problem 4.5 Relation of the two-slit experiment with polarizers to multiple Stern-Gerlach analyzer experiments. Write an essay that describes how the polarization experiments in Fig. 4.14 are related to the Stern-Gerlach analyzer-loop experiments with closable gates in Figs. 2.24, 2.23 and 2.26. Be sure to describe in your essay how the angles relate for these two different types of experiments.

Problem 4.6 Diagonal basis. Using the diagonal basis, given by

$$|D\rangle = \frac{1}{\sqrt{2}}(|V\rangle + |H\rangle) \quad \text{and} \quad |D'\rangle = \frac{1}{\sqrt{2}}(|V\rangle - |H\rangle), \quad (4.19)$$

we can write $|V\rangle = \frac{1}{\sqrt{2}}(|D\rangle + |D'\rangle)$ and $|H\rangle = \frac{1}{\sqrt{2}}(|D\rangle - |D'\rangle)$. Use this basis to analyze the experiment in Fig. 4.14(c). Be sure to show all of the work for each step in the state preparation.

Problem 4.7 The Quantum Mystery. Explain in your own words what the quantum mystery is, why the formalism does not directly address it, and why it cannot be explained using classical reasoning.

Problem 4.8 Conservation of probability in the Mach-Zehnder interferometer. Consider the case where the mirror on the sample path has been moved such that we have added a phase of θ to the sample path. Explicitly calculate the probability for detection in the bright and dark detectors and show that the sum of the two probabilities is 1 regardless of the value of θ .

Problem 4.9 State preparation for the dark detector. Redo the state preparation described in Eq. (4.16) for the state that goes to the dark detector.

Problem 4.10 Mach-Zehnder interferometer with polarizers. Write an essay describing what happens to the detection probabilities for the Mach-Zehnder interferometer when we have a V polarizer on the sample path and an H polarizer on the

reference path. Include in your discussion what happens when you add a polarizer at some angle θ before each of the detectors. This can be a qualitative discussion that does not need a quantitative analysis.

Problem 4.11 Delayed choice experiment on mach-Zehnder interferometers with polarizers Calculate probabilities at the bright detector for a polarizer at an angle that rotates from 0° (V) to 90° (H) for the case where we have a V polarizer on the sample path and an H polarizer on the reference path.

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